Two-Device Network Setup in Cisco Packet Tracer

Foundational networking lab

Name: EYITUOYO ATIGBI Date: 16 October, 2020.

Summary of project

As a step towards a career in cybersecurity, I built a small network in Cisco Packet Tracer connecting two computers, assigned them IP address, inspected their network details using ipconfig and ipconfig from the command lines, and tested connectivity with ping command.

Tools and Environment

Item	Details
Software	Cisco Packet Tracer
Devices	Two computers (Device A and Device B)
Cable	Copper cross-over cable
Interface	Fast Ethernet(100Mbps)
Addressing	Device A: 10.10.10.3, Device B: 10.10.10.4

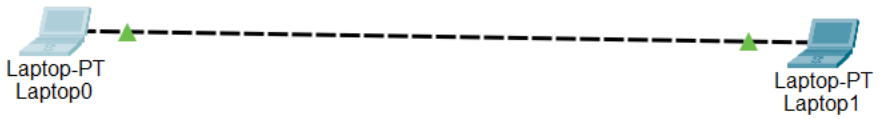
Steps Taken

1) Added two computers to the packet tracer workspace.



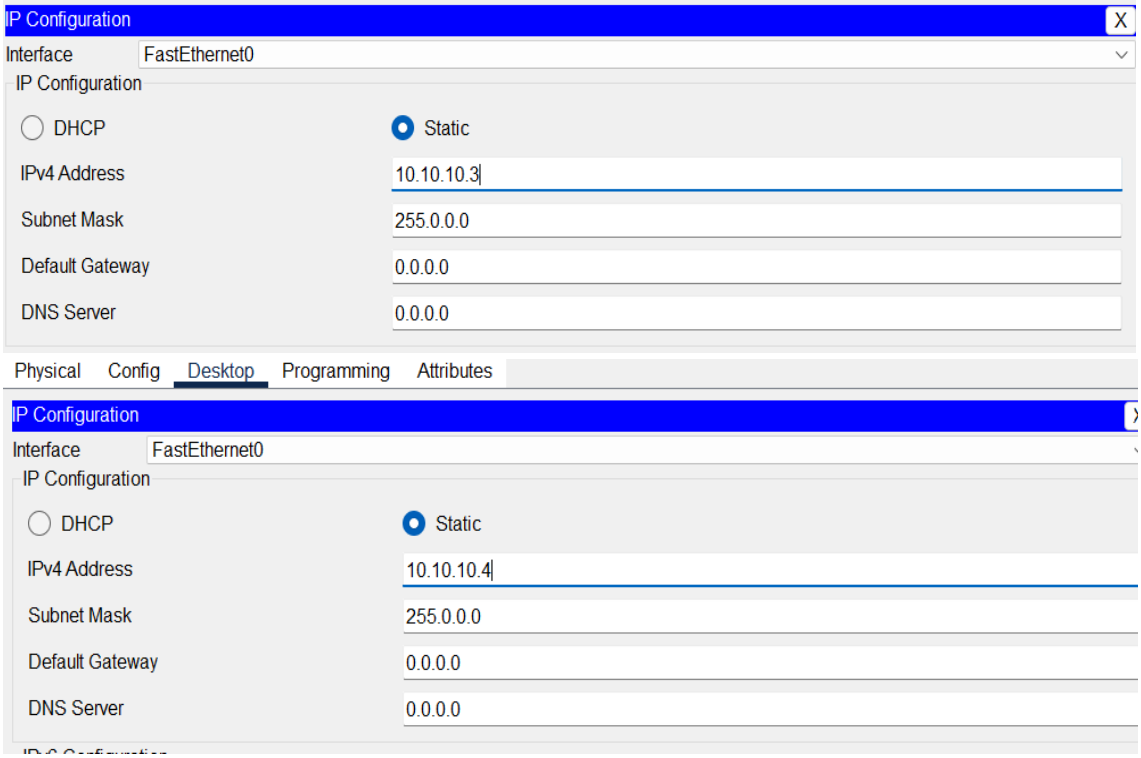


2) Connected them with a cross-over cable. A cross-over cable is used when connecting two similar devices directly (for example, computer to computer).

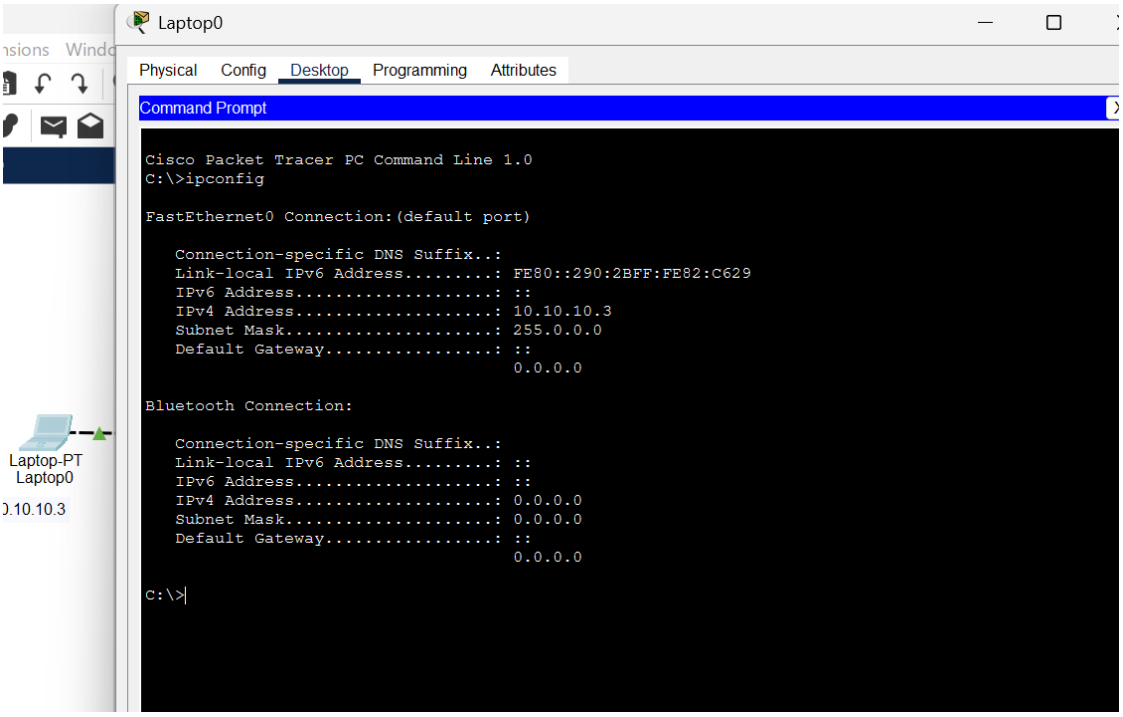


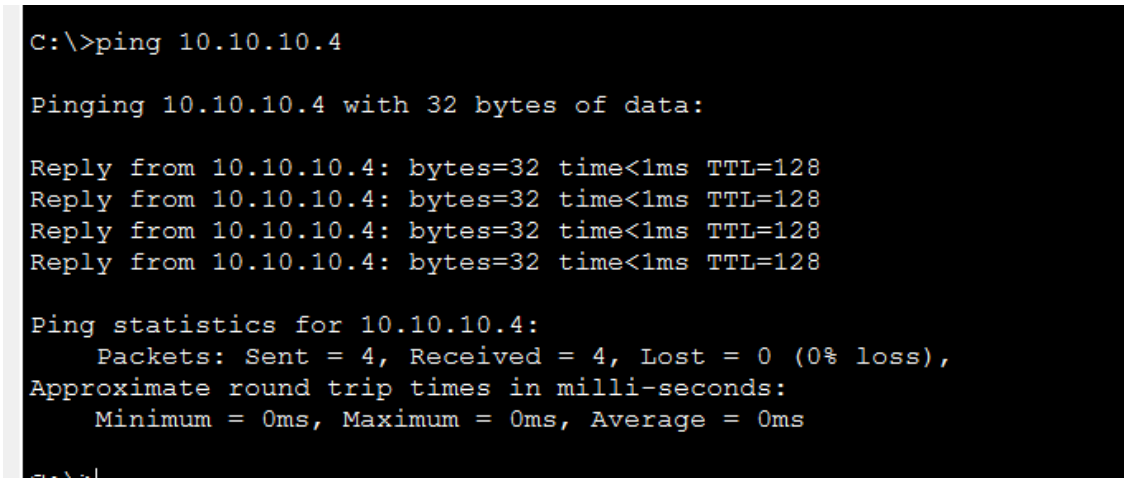
3) Selected the fast Ethernet port from the available options (RS-232, USB, Fast Ethernet). Choose Fast Ethernet because it is a real network connection that runs on TCP/IP and supports data sharing and remote tools.

4) Assigned IP address: 10.10.10.3 to Device A and 10.10.10.4 to Device B.



5) Ran commands such as ipconfig to ensure the ip address assigned is what shows, ipconfig /all to get the MAC address (physical address) and ping command to ensure communication is seamless.





Key Concepts Learned

1) Cross-over cable vs Straight through cable and when to use them. Cross-over cable is used for similar devices while straight through cable are used for different devices.
2) Fast Ethernet: a 100Mbps wired standard that uses TCP/IP.
3) IP address (logical, Layer 3) vs MAC address (physical, Layer 2).
4) ipconfig and ipconfig /all for viewing network configuration.
5) Ping as basic test for connectivity and troubleshooting.

Result

Both devices were configured successfully and ping replies confirmed that the connection was established.